

MAXWELL COLE, PHD

Medical Physics Resident | UC San Diego

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RESEARCH AND CLINICAL INTERESTS

My clinical interests are in developing treatment techniques and workflows that can be implemented with equipment and resources clinics already have, including streamlined approaches and automation of manual planning steps. My research interests are in computational and mathematical modeling of radiobiological and biophysical systems, an interest that began during my PhD studying vascular injury and whole-body blood flow at large scale. I am also interested in global health and scientific outreach, building open-access physics education programs and contributing to organizations focused on expanding access to medical care and physics training.

EDUCATION

Medical Physics Residency

Expected Completion: June 2027

Department of Radiation Medicine and Applied Sciences

University of California, San Diego

Ph.D. in Physics (Medical Physics)

2025

Louisiana State University

◊ Dissertation: “First Digital Biophysical Model of the Entire Human Cardiovascular System”

◊ Advisor: Wayne Newhauser, PhD

B.S. in Physics, Minor in Mathematics

2020

Louisiana State University

PROFESSIONAL CERTIFICATIONS

American Board of Radiology (ABR)

Part 1 – Completed August 2026 (*Results Pending*); Part 2 – Scheduled Summer 2027

CLINICAL TRAINING

Chief Resident

2026-Present

- Coordinated call schedules and task assignments for a cohort of 3 co-residents.
- Co-authored 3 departmental standard operating procedures.
- Served as resident representative to the Quality Committee, presenting safety incidents and developing corrective action strategies to prevent recurrence.
- Managed the incident-learning system and throughput tracking across the main site and 4 affiliate clinics.
- Served on the Communications Committee, managing the residency program’s Instagram account, staffing the residency recruitment booth at the AAPM Annual Meeting, and leading tours and information sessions for prospective applicants.
- Trained and mentored junior residents on clinical procedures, quality assurance workflows, and case planning.

Treatment Planning and Delivery

- Planned external beam radiotherapy cases spanning all major disease sites using the Eclipse treatment planning system (TPS).
- Planned and executed real-time replanning for online adaptive radiotherapy treatments using Ethos.
- Planned stereotactic radiosurgery/radiotherapy (SRS/SRT) treatments using both HyperArc and cone-based techniques.
- Planned single-field and multi-field optimized (SFO/MFO) proton therapy treatments.
- Planned MR-only, synthetic-CT-based prostate VMAT treatments.
- Utilized knowledge-based and automated planning tools to improve plan quality and efficiency.

Quality Assurance

- Performed TG-51 dose calibration on Varian TrueBeam, Ethos, and Trilogy linear accelerators.
- Performed routine and ad hoc machine QA on Varian TrueBeam, Ethos/Halcyon, and Trilogy linear accelerators; GE CT simulators; Varian Bravos HDR afterloader; MRI; and the Varian ProBeam proton gantry.
- Performed pre-treatment stereotactic radiosurgery quality assurance.
- Performed patient-specific quality assurance using portal dosimetry, log-file-based verification, and phantom-based measurements.

Commissioning and Accreditation

- Participated in commissioning of the AcurosXB dose calculation algorithm (v18) and the ARIA oncology information system upgrade (v18) (*see Clinical Projects*).

Clinical Responsibilities

- In-vivo dosimetry using MOSFET detectors.
- Electron insert factor measurements for custom-fabricated cutouts.
- One-on-one physics consults with patients and families on treatment plans and technical questions.
- EQD2 conversions for re-treatment cases.
- 4DCT simulation and motion management assessment for thoracic and abdominal cases.
- Secondary plan checks and weekly chart checks.

CLINICAL PROJECTS

Development and Analysis of a Virtual Cone Technique for Trigeminal Neuralgia Treatment with the Millennium MLC

Supervisor: Ryan Manger, PhD

- Optimized and generated a standardized MLC-based planning technique to reproduce the dose distribution of cone-based radiosurgery plans.
- Performed film-based dosimetric measurements to validate plan sphericity and conformity against cone-based benchmarks.
- Assessed the clinical feasibility of the technique against trigeminal neuralgia dose-specification criteria, identifying dynamic collimator rotation and reduced leaf-gap settings as the primary levers for improving dose sphericity.
- *Manuscript and conference presentation planned.*

Implementation of a Remote Viewer System for Physician-Covered Special Treatments

Supervisor: Ryan Manger, PhD

- Designed and built a Raspberry Pi-based device to mirror the treatment delivery system console and stream the session to a locally hosted address.
- Enabled covering physicians to remotely view on-treatment imaging, contours, and plans without being physically present at the machine console.

Development and Implementation of an Automated Electron Block Procedure

Supervisor: Ryan Manger, PhD

- Developed a program to automate treatment planning for custom electron cutout blocks.
- Reduced potential for error by eliminating manual insert-factor measurements and hand calculation of monitor units from the clinical workflow.

Additional Clinical & Technical Projects

- **MR-Only Simulation and Planning Workflow** – Assisted with faculty education and clinical implementation (*Supervisors: T. Pawlicki, PhD; R. Manger, PhD*)
- **AcurosXB v18 Dose Calculation Algorithm** – Assisted with commissioning and clinical implementation across Ethos, Halcyon, and TrueBeam (*Supervisor: R. Manger, PhD*)
- **ARIA v18 Upgrade** – Assisted with commissioning and clinical implementation (*Supervisor: R. Manger, PhD*)
- **Annual QA Program, Ethos and Halcyon** – Designed the annual test and measurement schedule for the clinic’s O-ring linear accelerators (*Supervisors: L. Padilla, PhD; J. Hosiak, PhD*)

RESEARCH EXPERIENCE

Resident Researcher

2025-Present

Department of Radiation Medicine and Applied Sciences, UC San Diego

- Managed 5 concurrent clinic-forward research projects spanning MLC-based radiosurgery technique development, treatment planning system and algorithm commissioning, and clinical workflow automation, prioritized to address emerging clinical needs (*see Clinical Projects*).
- Prepared and delivered educational materials for physician and physics faculty audiences.

Doctoral Researcher

2020-2025

Department of Physics & Astronomy, Louisiana State University

- Designed a prototype cardiovascular digital twin capable of modeling up to 34 billion vessels across the full human body using distributed supercomputing resources.
- Developed parallel-enabled biophysics models of blood flow, vessel deformation and rupture, and vasodilation using MPI and Kokkos runtime systems for exascale computing.
- Performed computational feasibility analyses of simulating radiation-induced vascular injury and hemodynamic changes at full-body scale, informing subsequent model development.
- Served as principal point of contact for a multi-institutional research collaboration spanning Los Alamos National Laboratory, Oak Ridge National Laboratory, and the RIKEN Center for Computational Science.
- Wrote and contributed to grant proposals, including two NSF proposals totaling over \$1.8 million in awarded funding (PI: Hartmut Kaiser).
- Mentored 3 undergraduate and 2 master’s students in research methods, data analysis, and scientific communication.

Department of Physics & Astronomy, Louisiana State University

- Developed personalized anthropomorphic phantoms for radiation dose measurement and quality assurance.
- Organized research meetings and facilitated cross-institutional research partnerships.
- Conducted literature reviews to support ongoing projects and inform research directions.

PUBLICATIONS & PRESENTATIONS

Manuscripts Submitted for Publication

1. **Cole, M**, Newhauser W, Diehl P, and Seleson P. “Computational feasibility and scalability of coupling peridynamic vessel-fracture simulation with a whole-body cardiovascular digital twin.” Submitted to *Biomechanics and Modeling in Mechanobiology*, August 2026.
2. **Cole, M**, Newhauser W, and Diehl P. “A multiscale biophysical model of nitric oxide diffusion and vasodilation: A computational feasibility study.” Submitted to *Biomechanics and Modeling in Mechanobiology*, August 2026.

Preprints

1. Newhauser W, **Cole, M**, Kaiser H, Tohid R, Chancellor J, Nader N, and Diehl P. “Toward Large-Scale Numerical Modeling of the Cardiovascular System with up to 34 Billion Vessels.” *bioRxiv* 2026.05.22.727287 (2026).

3 additional manuscripts in preparation; titles and details available upon request.

Invited Talks & Conference Presentations

1. **Cole, M**. *First Digital Biophysical Model of the Entire Human Cardiovascular System* [Invited Talk]. Advances in Applied Computer Science Invited Speaker Series, Los Alamos National Laboratory, Los Alamos, NM. August 27, 2025.
2. **Cole, M**. *Simulating Space Radiation-Induced Changes to Cardiovascular Blood Flow* [Poster]. Aerospace Medical Association Annual Scientific Meeting, New Orleans, LA. May 25, 2023.
3. **Cole, M**. *Computational Feasibility of Simulating Radiation-Induced Changes in Vasculature and Blood Flow Rates in the Entire Human Body* [Poster]. Workshop on Asynchronous Many Task Applications, Baton Rouge, LA. February 15, 2023.

TEACHING EXPERIENCE

Joint Conference Case Lecture

2026

UC San Diego

- Presented a lecture on the radiation treatment of trigeminal neuralgia.
- Collaborated with a radiation oncology resident to co-develop and deliver the lecture to clinical faculty.

Resident Lecture Series

2025-2026

UC San Diego

- Presented a series of lectures to physics faculty covering:
 - Proton QA
 - MR-Only Simulation and Planning
 - Commissioning and QA of Treatment Planning Systems
 - TG-51
 - Biennial QA for Halcyon and Ethos

- Treatment Planning Considerations for Gastrointestinal SBRT
- TG-29: Total Body Irradiation

Guest Lecturer

2022

Louisiana State University

- Presented a 5-lecture course module on research methods and technologies to graduate students.
- Designed the module curriculum and delivered real-time coding demonstrations.
- Prepared in-class and take-home assignments to assess student comprehension.

Teaching Assistant / Learner's Assistant

2019-2021

Louisiana State University

- Delivered lectures, held office hours, and graded assignments for undergraduate physics courses (2020-2021).
- Tutored underclassmen and led review sessions to reinforce course material (2019-2020).

HONORS & AWARDS

Dr. Charles M. Smith Superior Graduate Student Scholarship; \$20,000

2024

Louisiana State University

Roussel Family Graduate Student Award in Communication; \$2,000

2023

Louisiana State University

Best Poster Award

2023

Workshop on Asynchronous Many Task Applications

Kenneth R. Hogstrom Superior Graduate Student Scholarship Award; \$20,000

2022

Louisiana State University

LaSPACE Graduate Student Research Award; \$16,000

2022

Louisiana Space Grant Consortium

Inspire Award for Exemplary Ethical Action; \$1,000

2020

LSU Ethics Institute

OUTREACH AND VOLUNTEER WORK

Project Co-Lead, Medical Physics Brain (Rayos Contra Cancer)

2026-Present

- Lead and manage an effort to build an open-access medical physics education repository.
- Consult with and recruit practicing medical physicists to contribute clinical and technical content.
- Design and architect the platform's structure and content organization.

Advocate/Judge, Louisiana Science & Engineering Fair, Region VII

2022

- Judged science projects for middle and high school students.
- Served as a scientific advocate, providing students with resources and feedback to advance their projects.

Co-Founder, One Louisiana NOW!

2020

- Co-founded a non-profit relief organization to supply personal protective equipment to hospitals in need during the COVID-19 pandemic.
- Researched and designed prototype protective gear based on hospital-identified needs.
- Coordinated with institutions and hospital staff to ensure timely delivery of supplies.

- Co-authored an unpublished internal report, “A Community-Based Response to Equipment Shortages Caused by the COVID-19 Pandemic,” circulated among participating hospitals and physicians; the report was subsequently used by other organizations as a model to establish their own PPE response programs.

SERVICE

Professional Organizations

- American Association of Physicists in Medicine (AAPM) 2019-Present
- Radiation Research Society (RRS) 2022-Present

Journal Peer Reviewer

- *Medical Physics*, AAPM 2026
- *PLOS One* 2025

Institutional Committees

- Quality Committee, UC San Diego 2026-Present
- Communications Committee, UC San Diego 2026-Present

TECHNICAL SKILLS

Treatment Planning & Oncology Information Systems: Eclipse, Ethos, ARIA, ESAPI

Programming Languages: Python, C, C#, C++, MATLAB, Swift

Software & Systems Development: iOS app development (Swift/SwiftUI), web development, Git/version control, embedded systems (Raspberry Pi)

HPC Systems: scripting, job management systems, parallel computing (*e.g.*, MPI, Kokkos)

Data Visualization: matplotlib, gnuplot, ParaView